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Matrix Correction

The contents of the water (matrix) have an important influence to the ISE membranes. The membranes need some hours to adapt to the matrix.

To compensate the influences from matrix it is recommended to perform a matrix correction by a reference analysis in lab.

In the calibration menu you can store the value at the time of sampling.

If you find significant difference to the value the probe shown at time of sampling, you can adapt your probe to the wastewater matrix by enter your lab value in the calibration menu.

With this **Matrix Correction** **the probe is adapted to your wastewater application perfectly!**

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Matrix Correction - Overview

Table 1 Correction options for the NO3D sc sensor

Correction Option	Application
MATX1	The most commonly used Matrix correction. It performs a 1 point matrix correction (offset adjustment) for nitrate (4.6.3.1 on page 24).
MATX1 Cl-	With the MATX1 Cl- the nitrate value will be corrected as with the MATX1 and in addition the chloride value can be corrected. This is necessary if a higher precision of the nitrate value is required as the chloride value will interfere the nitrate measurement (4.6.3.2 on page 24).
MATX2	Where there is a dynamic process with a large nitrate fluctuation (at least 1/2 decade between the lowest and highest concentration)[1], it is recommended to perform the MATX2 (4.6.3.3 on page 25).
MATX2 Cl-	Where there is a dynamic process with a large nitrate fluctuation (at least 1/2 decade between the lowest and highest concentration)[1] and chloride value is to be corrected, it is recommended to perform the MATX2 Cl- (4.6.3.4 on page 26).
VALUE CORR	The nitrate and chloride displayed value and the nitrate laboratory value can be entered for two points. This is a different way to perform a MATX2 Correction. Here, for two points, the displayed nitrate and chloride values and the nitrate laboratory value are entered via an input screen.
PREVIOUS CAL	Activation of one of the last 4 matrix and value corrections performed.
FACTORY CAL	If the current sensor code is no longer available, average data for the sensor can be activated using FACTORY CAL.

[1] Example of a half decade: The nitrate nitrogen concentration moves between 1 and 5 mg NO_3–N or between 5 and 25 mg/L NO_3–N.

Note: Calibration points should be higher than 1 mg/L if possible

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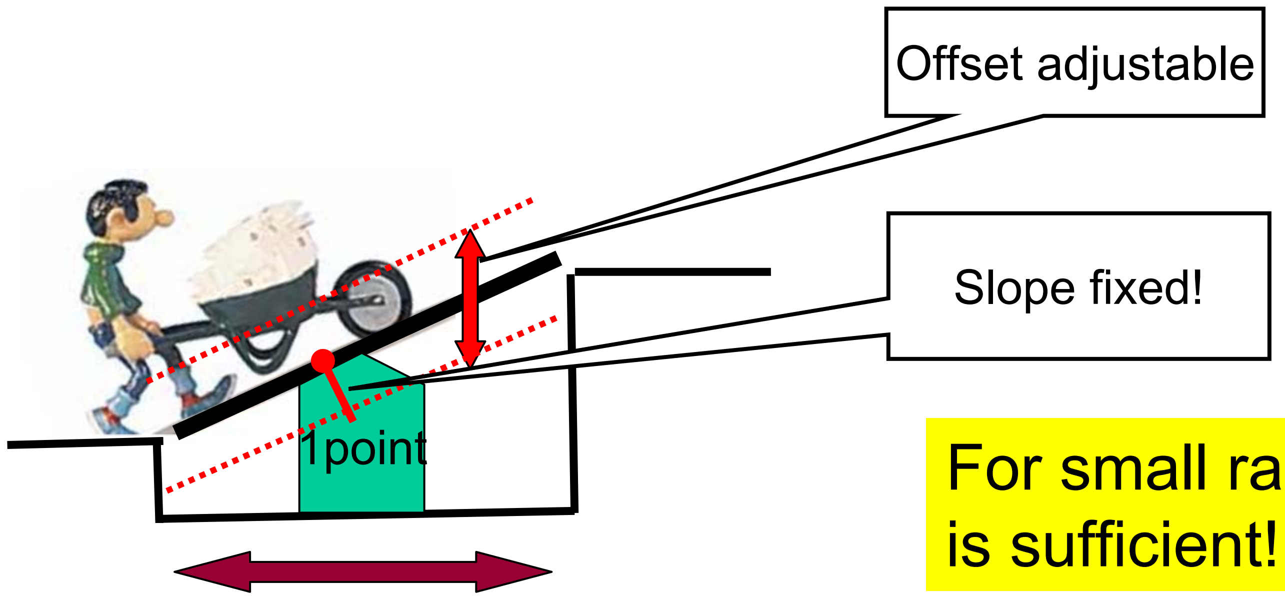
Matrix Correction – 1 Point Correction

Should be done (earliest) 12 hours after installation of the sensor cartridge. Membranes need some hours for adaption to the wastewater.



Small Range / less changes in measured value

For small range 1 point is sufficient!

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Matrix Correction – 1 Point Correction

4.6.3.1 MATX1 correction (one point matrix correction)

Proceed as follows to perform the MATX1:

CALIBRATE
CAL. CONFIG.
DATE
CONC MEAS 1
SET NO3–N CONC

1. Select **SENSOR SETUP>NO3D SC>CALIBRATE> CAL CONFIG**.
2. Select **MATX1** in the selection window and press **ENTER**.
3. Select **CONC MEAS 1**.

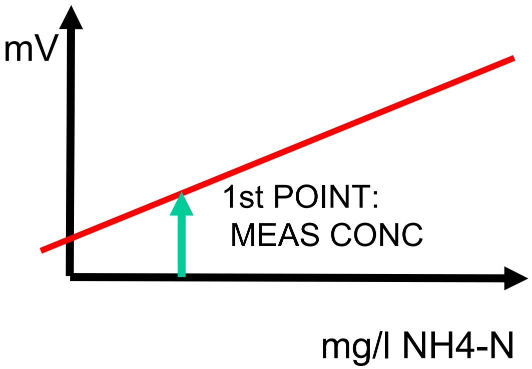
READING STABLE?
NO3–N:
DRIFT
Cl-
DRIFT

The currently measured nitrate and chloride values are displayed. The drift indicates whether the measured value is stable.

4. Wait until the measured value is stable and confirm by pressing **ENTER** (Drift should be < 0.03 mg/L). The values for nitrate and chloride are saved.
5. Immediately after saving, take a water sample for laboratory analysis from as close to the sensor as possible.
6. Analyze the sample immediately after taking the sample as its nitrate content can change quickly.

After determining the laboratory reference value, proceed as follows:

7. Select **SENSOR SETUP>NO3D SC>CALIBRATE> SET NO3–N CONC**.
8. Enter the laboratory value (reference value) for NO_3–N and press **ENTER** to confirm. Confirmation of the entered laboratory value activates the matrix correction.



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Matrix Correction – 1 Point Correction with Chloride

4.6.3.2 MATX1 Cl- correction

Proceed as follows to perform the MATX1 Cl-:

CALIBRATE
CAL. CONFIG.
DATE
CONC MEAS 1
SET Cl- CONC
SET NO3–N CONC

1. Select **SENSOR SETUP>NO3D SC>CALIBRATE> CAL CONFIG**.
2. Select **MATX1 CL-** in the selection window and press **ENTER**.
3. Select **CONC MEAS 1**.

READING STABLE?
NO3–N:
DRIFT
Cl-
DRIFT

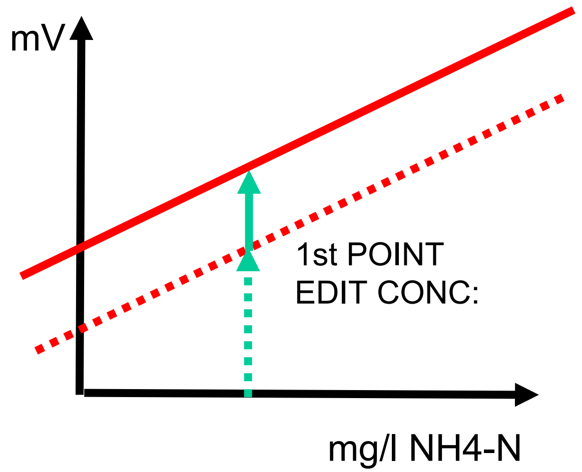
The currently measured nitrate and chloride values are displayed. The drift indicates whether the measured value is stable.

4. Wait until the measured value is stable and confirm by pressing **ENTER** (Drift should be < 0.03 mg/L). The values for nitrate and chloride are saved.
5. Immediately after saving, take a water sample for laboratory analysis from as close to the sensor as possible.
6. Analyze the sample immediately after taking the sample as its nitrate content can change quickly.

After determining the laboratory reference values for chloride and nitrate, proceed as follows:

7. Select **SENSOR SETUP>NO3D SC>CALIBRATE> SET NO3–N CONC**.
8. Enter the laboratory reference value for NO_3–N and press **ENTER** to confirm.
9. Select **SENSOR SETUP>NO3D SC>CALIBRATE> SET CL- CONC**.
10. Enter the laboratory reference value for chloride and press **ENTER** to confirm.

    Confirmation of the entered laboratory value activates the matrix correction.



Note:
For the chloride reference value we recommend to use the Quantab Chloride-Teststrips (33 to 649 mg/L Cl). You can order it with the order number 27449-40

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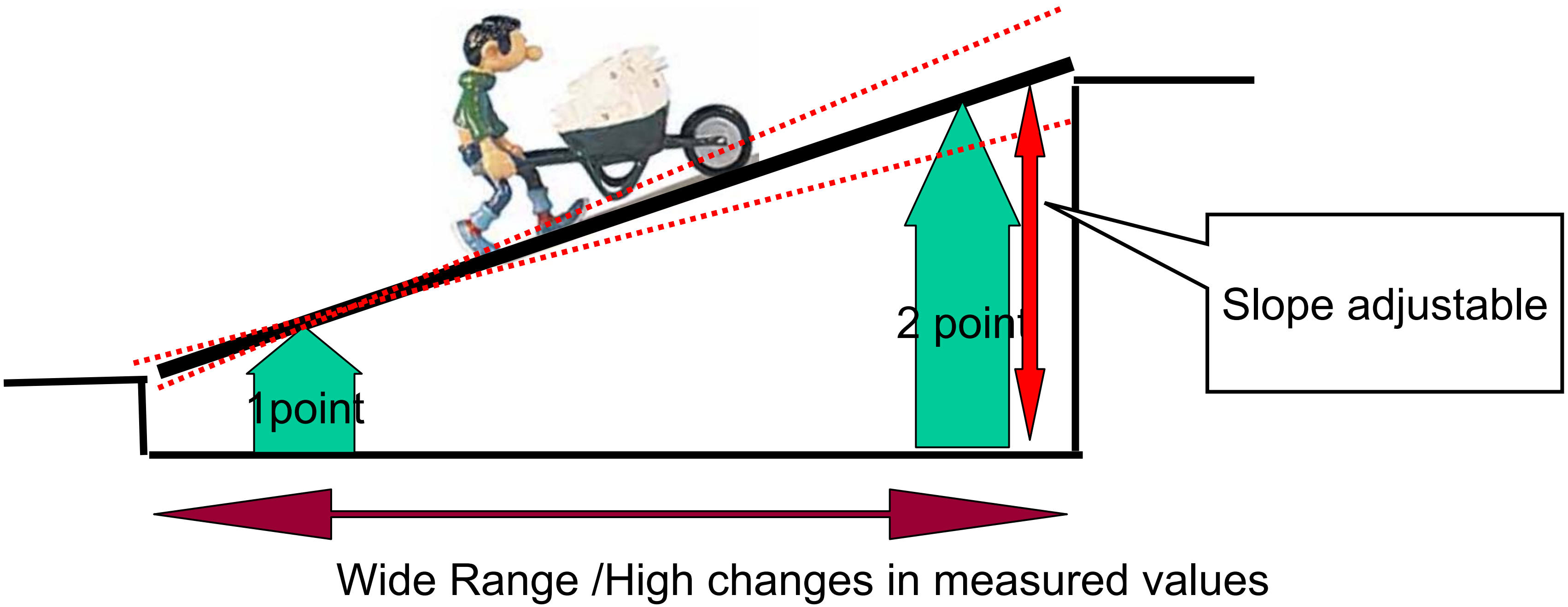
Matrix Correction – 2 Point Correction

2 Point Matrix Correction should be used in dynamic processes where the nitrate value fluctuates at least ½ decade (e.g.: 2 and 10 mg/L)



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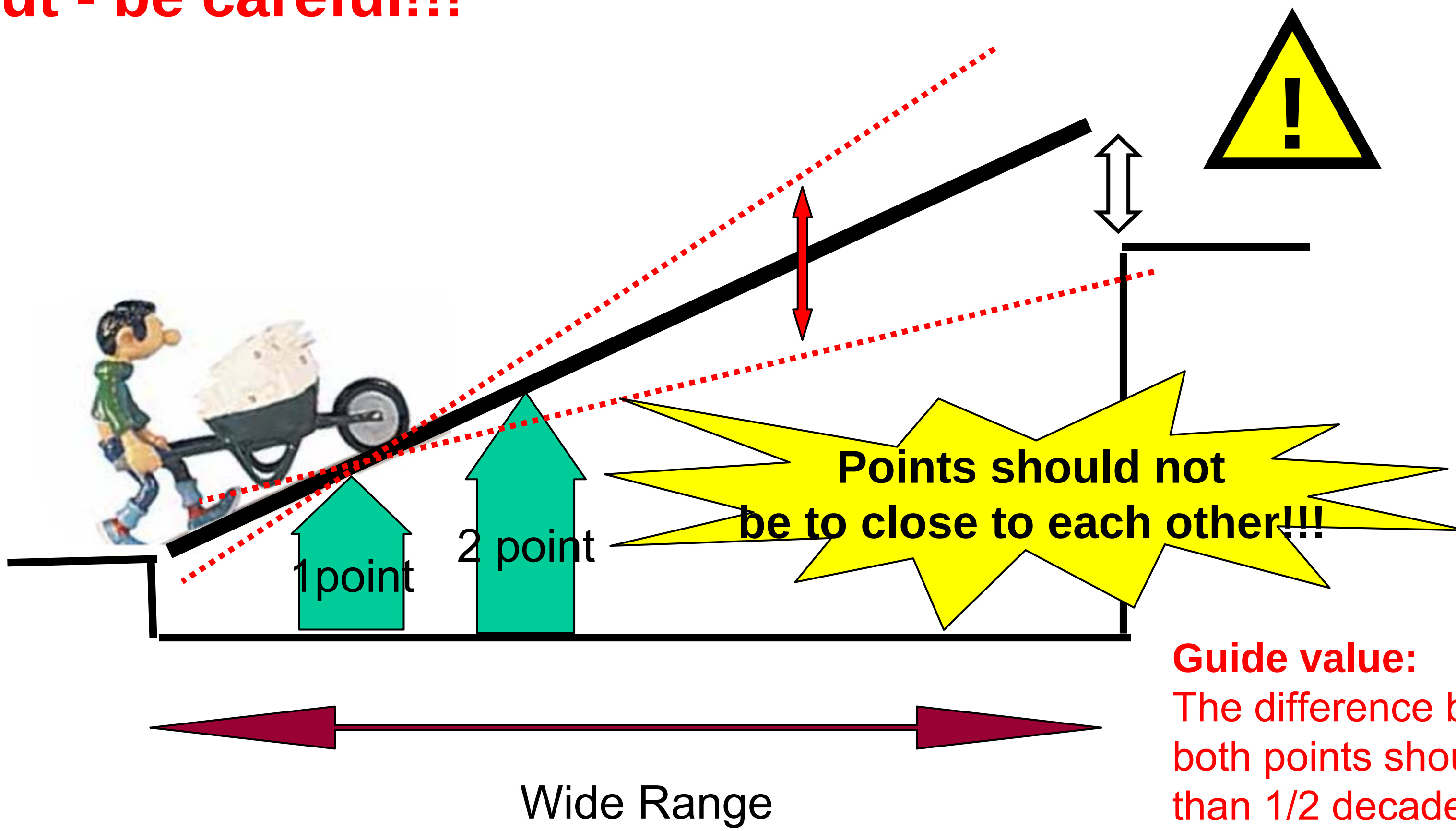
Matrix Correction – 2 Point Correction

But - be careful!!!



Guide value:
The difference between both points should be more than 1/2 decade

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Matrix Correction – Value correction

The value correction offers the possibility to enter a correction according to comparison measurements which have been performed in the past.
To perform the value correction you should take several samples at different times and concentrations and analyse them in the lab. At the same time when you grab the sample you should note the nitrate and the chloride value of the NO3Dsc.
As a template you can use therefore the following table:

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Matrix Correction – Value correction

Date	Time	NO3-N NO3Dsc value	CL^- NO3Dsc value	NO3-N Lab value

Choose afterwards two points with the biggest difference (at least a ½ decade) and the most plausible results and enter this into the sensor software.
With the value correction you have performed a 2 point matrix correction.
See the following section from the manual on how to perform the value correction.

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Matrix Correction – Value correction

4.6.3.5 Value correction

Value correction offers the option of subsequently correcting a matrix at two different concentrations.

Take several samples on various days with different concentrations and perform an analysis of the samples in the laboratory.

Note: *The concentrations should be in a concentration range of at least half a decade.*

1. Note the two displayed nitrate and chloride values, which are measured with the sensor at the time of the sampling.
2. Then note the measured nitrate laboratory reference value.

All 3 values form a correction point.

3. From the values recorded, select the two correction points with nitrate concentrations as far apart as possible.
4. Select **CAL CONFIG.> CALIBRATE>VALUE CORR** from the menu and press **ENTER** to confirm.

Enter the 3 noted values:

5. **1. NO3–N NO3D sc:** Enter the displayed NO_3–N value for the 1st correction point. Press **ENTER** to confirm.
6. **1. Cl- NO3D sc:** Enter the displayed Cl^- value for the 1st correction point. Press **ENTER** to confirm.
7. **1. NO3–N lab:** Enter the measured NO_3–N laboratory reference value for the 1st correction point. Press **ENTER** to confirm.
8. **2. NO3–N NO3D sc:** Enter the displayed NO_3–N value for the 2nd correction point. Press **ENTER** to confirm.
9. **2. Cl- NO3D sc:** Enter the displayed Cl^- value for the 2nd correction point. Press **ENTER** to confirm.
10. **2. NO3–N lab:** Enter the measured NO_3–N laboratory reference value for the 2nd correction point. Press **ENTER** to confirm.

    Value correction is activated and displayed as **MATX2**.

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Interferences

NO2-N:	1 mg/L NO2-N ≈ 0.2 mg/L NO3-N
Cl^-:	1 mg/L Cl^- ≈ 0.004mg/L NO3-N
Detergents:	Anionic detergents shows higher NO3-N results when they are over a certain time in front of the membrane (detergents report is pending)